

2015 MJC Prelim H2 Paper 3 Suggested Answers



(ii) Entropy is a measure of degree of disorder in a system.

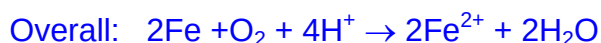
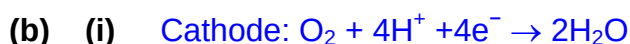
(iii) Entropy change is positive as the reaction shows an increase of number of moles of gas. Hence, the system is more disordered.

(iv) $\Delta H = (0.75)(-393.51) - (0.5)(-824.25) = +117 \text{ kJ mol}^{-1}$

(v) At low temperatures, positive ΔH outweighs negative $-T\Delta S$ or $|\Delta H| > |-T\Delta S|$. Hence ΔG is positive / > 0 , reaction is not feasible.

$$117 - T \left(\frac{140}{1000} \right) \leq 0$$

$$T \geq 836 \text{ K}$$



(ii) $E^\ominus_{\text{cell}} = +1.67 \text{ V}$

(iii) The increase in $[\text{H}^+]$ causes the equilibrium position of $\text{O}_2/\text{H}_2\text{O}$ to shift to the right making $E^\ominus_{\text{red}}$ more positive. Hence, overall $E^\ominus_{\text{cell}}$ to be more positive.

(iv) $E^\ominus_{\text{Zn}^{2+}/\text{Zn}} = -0.76 \text{ V}$

$E^\ominus_{\text{Fe}^{2+}/\text{Fe}} = -0.44 \text{ V}$

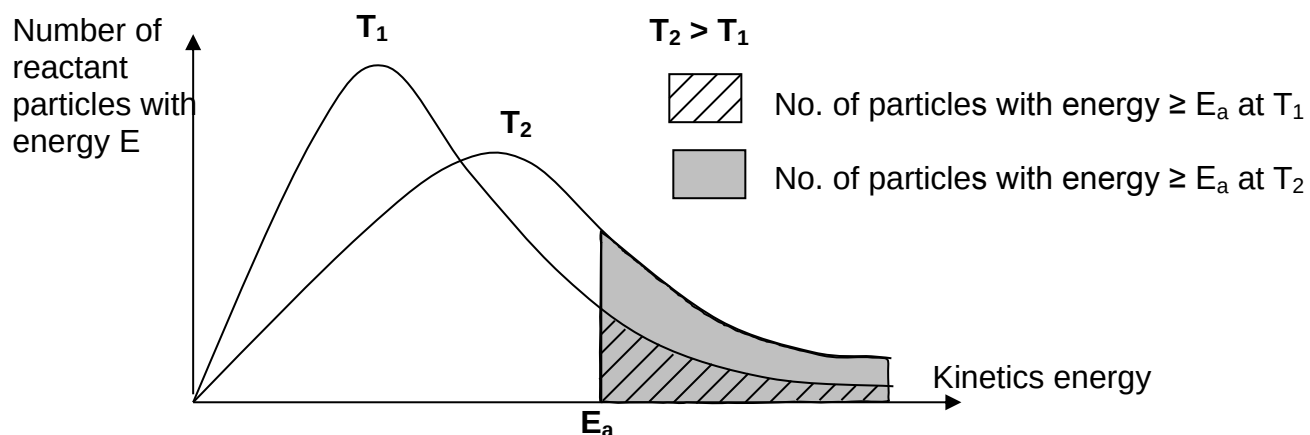
As Zn is preferentially oxidised and is a better reducing agent than Fe as $E^\ominus_{\text{Zn}^{2+}/\text{Zn}}$ is more negative.

(v) High activation energy, causing the process to be slow.

(c) (i) $3.16 \times 10^{-5} = k(0.4)$ hence $k = 7.90 \times 10^{-5} \text{ min}^{-1}$

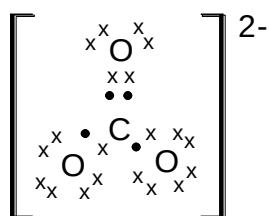
$$t_{1/2} = \frac{\ln 2}{k} = \frac{\ln 2}{7.90 \times 10^{-5}} = 8.77 \times 10^3 \text{ min}$$

(ii)



When temperature is increased, the average kinetics energy of reactant particles increases. The number of reactant particles with energy $\geq E_a$ increases leading to increase in frequency of effective collisions.

2(a)(i)



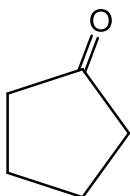
(ii) Number of mole of $\text{CaCO}_3 = \frac{1.12}{100.1} = 0.01119$

Mass of pure $\text{Ca}(\text{CH}_3\text{CO}_2)_2 = 0.01119 \times 158.1 = 1.769 \text{ g}$

Percentage purity = $\frac{1.769}{2.00} = 88.5 \%$

(iii) CaO and CO_2 .

(b) (i)



(ii) Ability of cation to polarise electron cloud of anion and break C–C bond increases. Compound is less thermally stable. Hence, thermal decomposition temperature decreases.

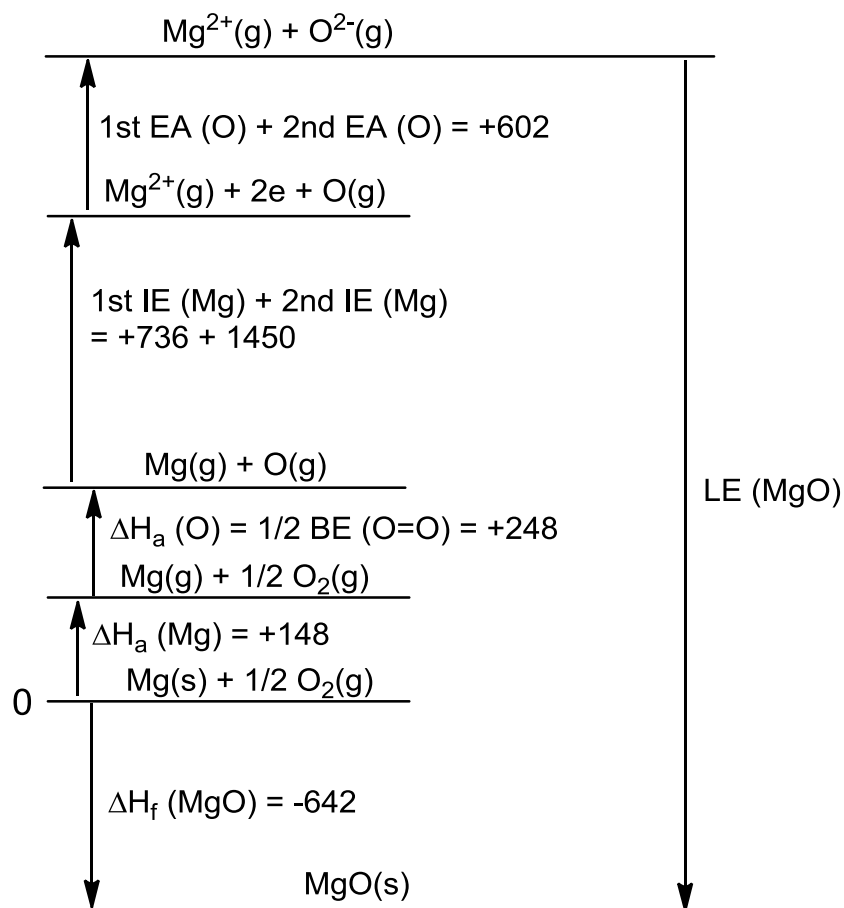
(c)(i) Energy evolved when one mole of ionic compound is formed from its gaseous ions under standard conditions .

(ii) Giant ionic lattice structure

$$|LE| \propto \frac{q_+ \cdot q_-}{r_+ + r_-}$$

- ionic radius: $\text{CO}_3^{2-} > \text{O}^{2-}$
- strength of ionic bonds: $\text{MgCO}_3 < \text{MgO}$
- energy required to overcome strong electrostatic force: $\text{MgCO}_3 < \text{MgO}$

(iii)

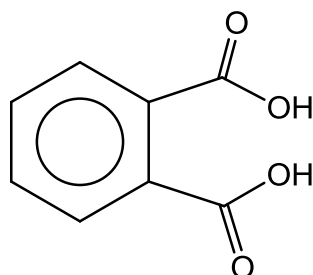


$$-642 = +148 + +248 + 736 + 1450 + 602 + LE(\text{MgO})$$

$$LE(\text{MgO}) = -3826 \text{ kJ mol}^{-1}$$

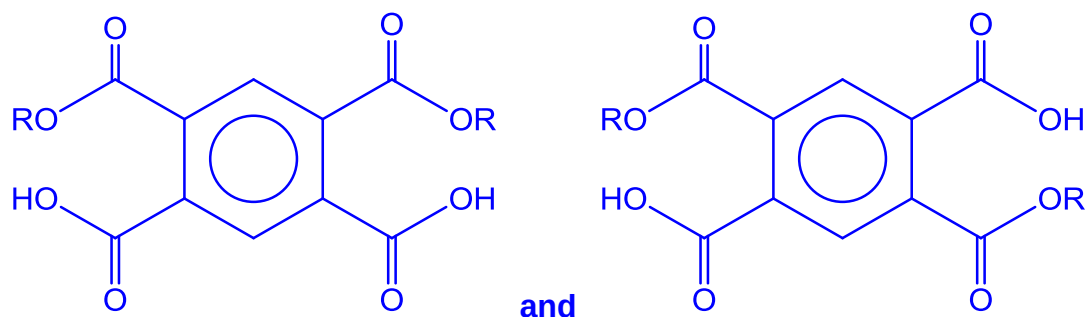
- (iv) The partial covalent character due to the polarization of the large CO_3^{2-} electron cloud by Mg^{2+} .

(d)(i)



(ii) 1. nucleophilic substitution

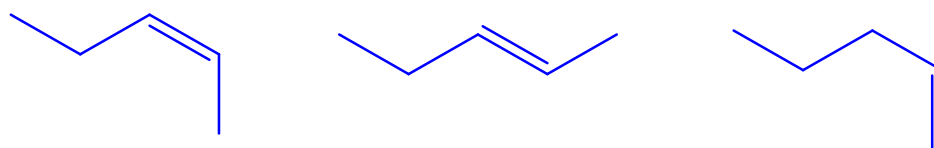
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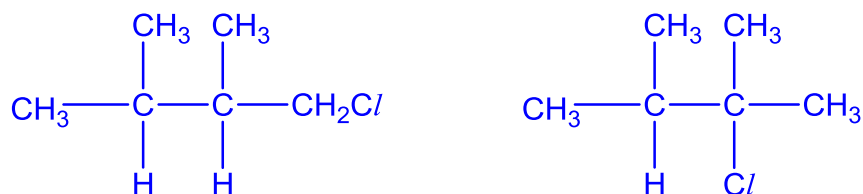
(iii) Generate RO^- which is a stronger nucleophile. Hence, nucleophilic substitution takes place at a faster rate.

3(a)(i) Electrophile.

(ii)



(iii)



Ratio is 2 : 7

(b) (i) Amount of heat absorbed by water = 41.8 kJ

$$\text{Amount of heat evolved by combustion} = \frac{41.8}{\frac{80}{100}} = 52.25 \text{ kJ}$$

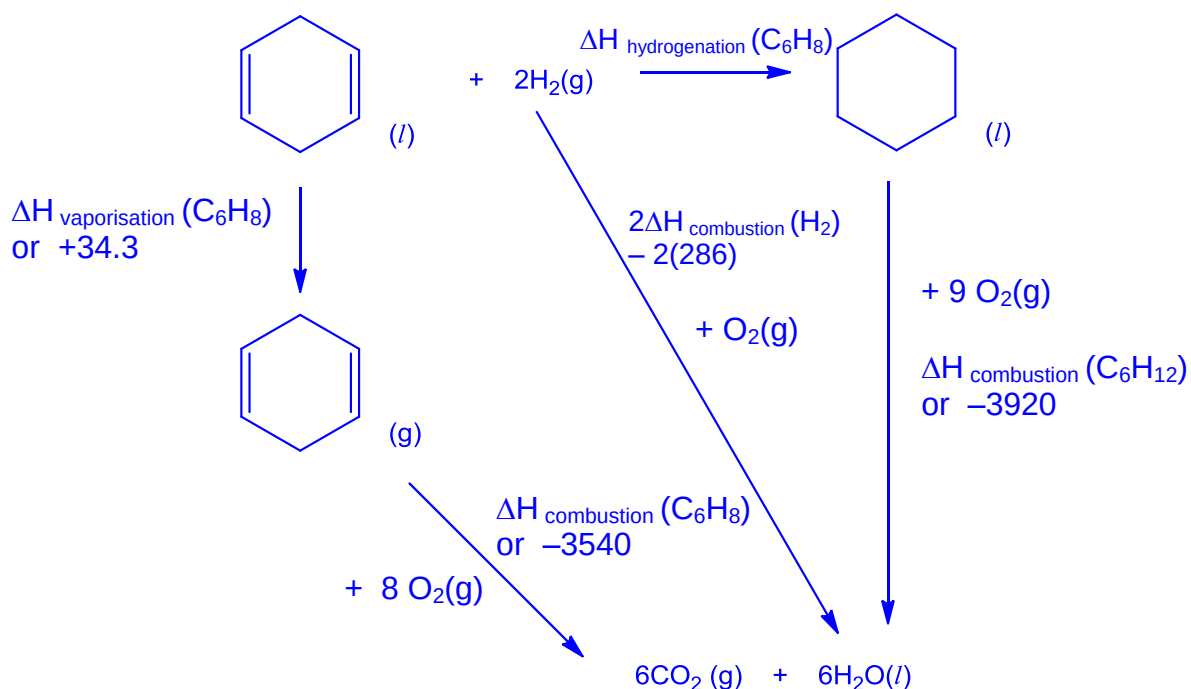
$$\text{No. of moles of cyclohexane required} = \frac{52.25}{3920} = 0.01333 \text{ mol}$$

$$\text{Mass of cyclohexane required} = 0.01333 \times [6(12.0) + 12(1.0)] = 1.12 \text{ g}$$

(ii) This is because energy is required to convert the ice to water is required.

$$(iii) \frac{1}{84} \times 3920 = 46.7 \text{ kJ g}^{-1}$$

(iv)



$$\Delta H_{\text{rxn}} = 34.3 + (-3540) + 2(-286) + 3920 = -157.7 \text{ kJ mol}^{-1}$$

- (c)(i) $2\text{H}_2\text{O} (\text{l}) + 2\text{e}^- \rightarrow \text{H}_2 (\text{g}) + 2\text{OH}^- (\text{aq})$
 The OH^- turns phenolphthalein pink.
 $2\text{CH}_3\text{CO}_2^- (\text{aq}) + 2\text{H}_2\text{O} (\text{l}) \rightarrow \text{C}_2\text{H}_6 (\text{g}) + 2\text{CO}_2 (\text{g}) + \text{H}_2 (\text{g}) + 2\text{OH}^- (\text{aq})$

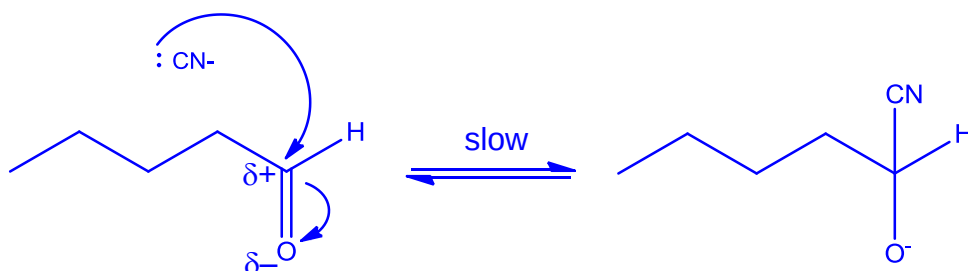
(ii) No of moles of ethane produced = $\frac{3.60}{24.0} = \frac{1 \times 15 \times 60}{2 \times 96500}$

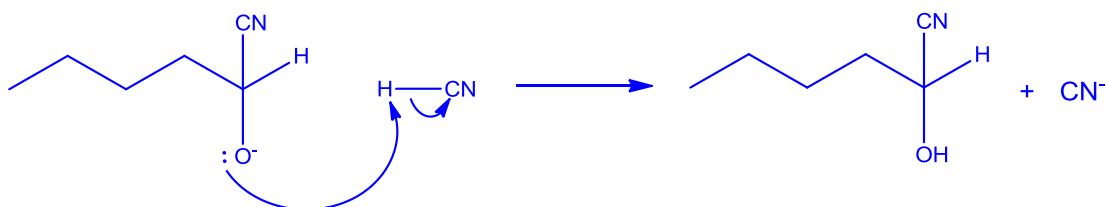
Current required = 32.2 A

Total volume of gases produced = $3.6 + 3.6 \times 2 + 3.6 = 14.4 \text{ dm}^3$

- (iii) For an ideal gas, there should be negligible intermolecular forces of attraction. C_2H_6 will behave more like an ideal gas. Since the size of the electron cloud of C_2H_6 is smaller than CO_2 , it has weaker and less extensive intermolecular van der Waals' forces of attraction than CO_2 .

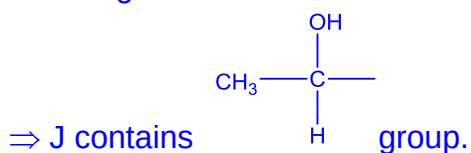
4(a) Nucleophilic addition





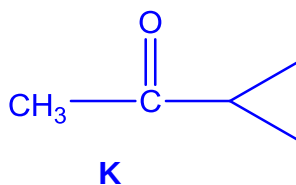
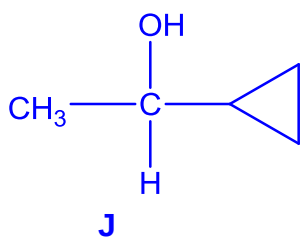
(b) J undergoes acid-metal reaction with Na $\Rightarrow$ J contains an alcohol group.

J undergoes oxidation with alkaline iodine solution



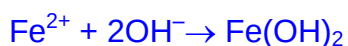
J undergoes oxidation with $\text{KMnO}_4 \Rightarrow$ J does not contain an alkene group.

K undergoes condensation with 2,4 DNPH. $\Rightarrow$ K is a ketone and J is a 2° alcohol.

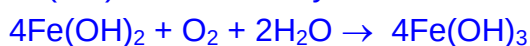


(c)(i) +2

When aqueous NaOH is added to solution A containing, a green precipitate, $\text{Fe}(\text{OH})_2$ is observed.



$\text{Fe}(\text{OH})_2$ is oxidised by air to red-brown $\text{Fe}(\text{OH})_3$.



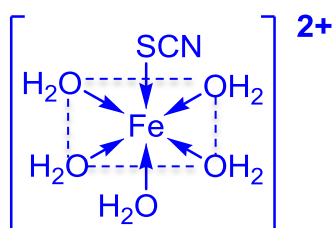
(ii) $2\text{Fe}^{2+} + \text{Cl}_2 \rightarrow 2\text{Fe}^{3+} + 2\text{Cl}^-$

When chlorine gas is bubbled into solution A, a yellow solution B is obtained.



$$E^\theta_{\text{cell}} = 1.36 - (0.77) = +0.59 \text{ V} > 0 \Rightarrow \text{reaction is feasible}$$

(iii) Ligand exchange



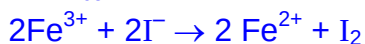
(iv)

(v) Fe^{3+} ions have partially filled 3d orbitals. d orbitals in Fe ions are split into two groups of different energy levels. The d electron undergoes d-d transition and is promoted to the higher d orbital. During the transition, the d electron absorbs a wavelength of light from the green region of the electromagnetic spectrum and the remaining wavelengths not absorbed appears as the resulting red colour.

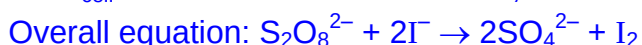
(vi) Fe^{2+} functions as a homogeneous catalyst.



$$\Rightarrow E^\ominus_{\text{cell}} = 2.01 - 0.77 = +1.24 \text{ V} > 0, \text{ reaction is feasible.}$$



$$\Rightarrow E^\ominus_{\text{cell}} = 0.77 - 0.54 = +0.23 \text{ V} > 0, \text{ reaction is feasible.}$$

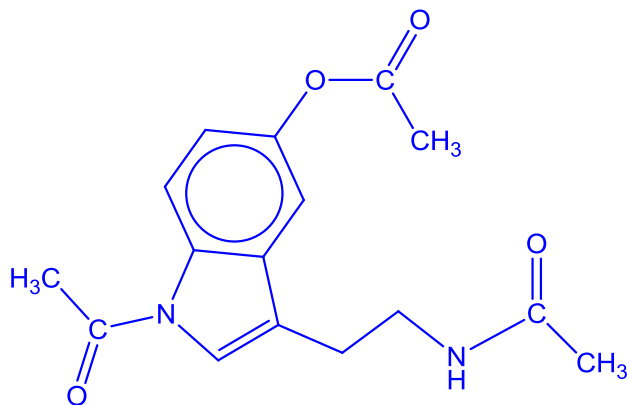


Oppositely charged ions are involved and attract each other. Activation energy is lower and thus reaction is faster.

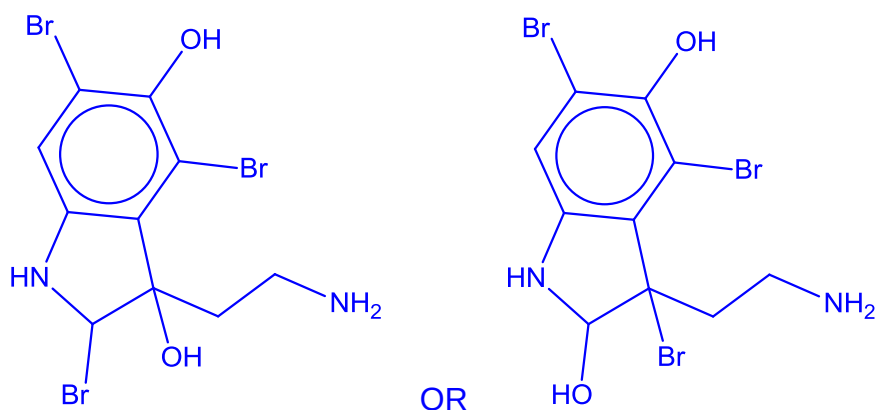
5 (a)(i) The electron-donating alkyl group increases the electron density of the lone electron pair on N_x atom of Serotonin. Hence, the lone electron pair on N_x atom is more available to accept a proton. For N_y atom of Serotonin, the lone pair of electrons on the nitrogen atom is delocalised into the benzene ring and $\text{C}=\text{C}$, thus the lone electron pair on N_y atom is less available to accept a proton.

(ii)

1.

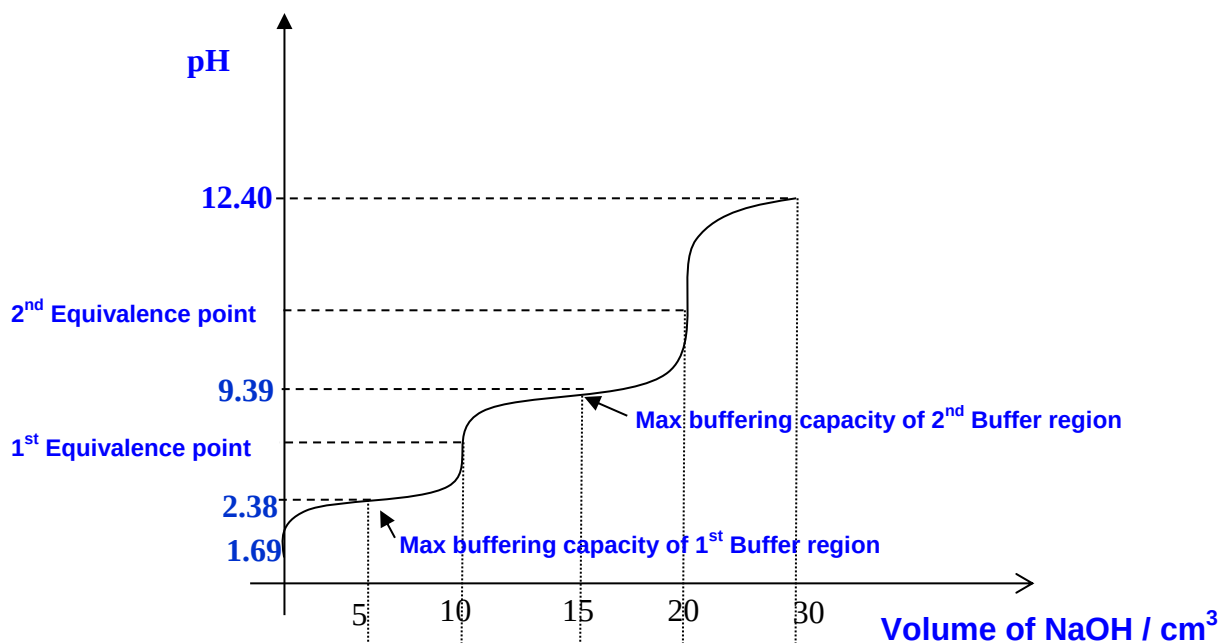


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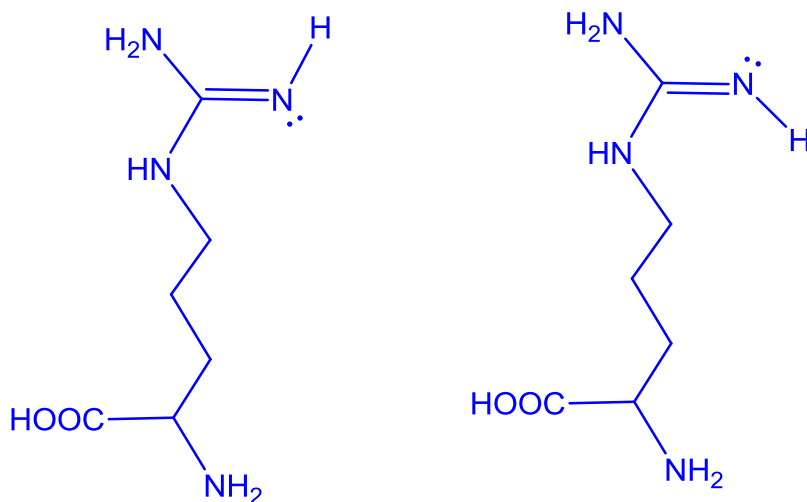


(b)(i) $[H^+] = \sqrt{(0.10 \times 10^{-2.38})} = 0.0204 \text{ mol dm}^{-3}$
Initial pH = 1.69

(ii) $[OH^-]_{\text{excess}} = 1.00 \times 10^{-3} / (40/1000) = 0.0250 \text{ mol dm}^{-3}$
 $pOH = -\lg 0.025 = 1.60$ so $pH = 14 - 1.60 = 12.40$



(c)(i)



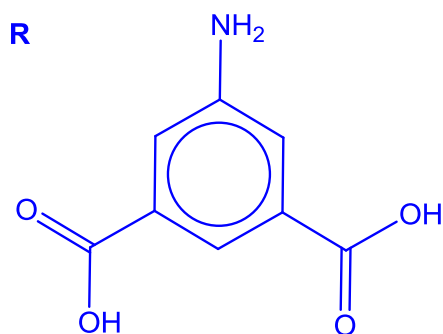
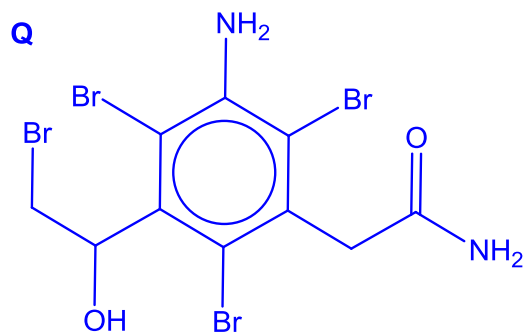
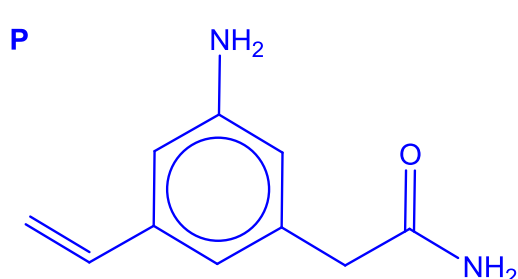
(ii) 8

(d) P does not undergo neutralisation with NaOH (aq) but it undergoes neutralisation with HCl (aq) hence P does not contain phenol group but contains amine.

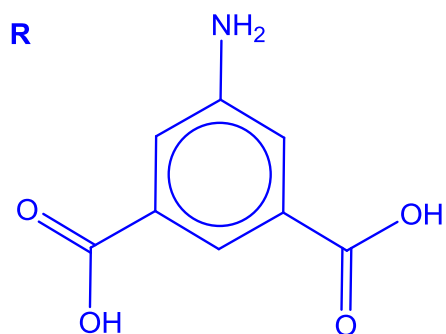
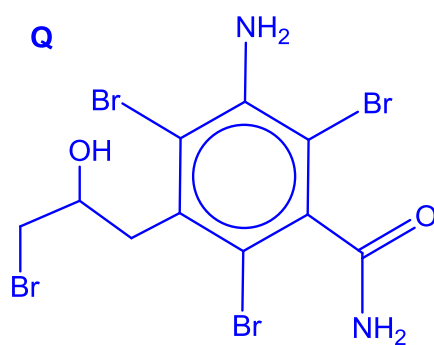
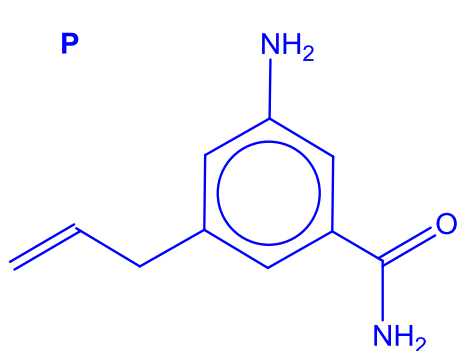
P undergoes basic hydrolysis with NaOH (aq) to form ammonia gas hence P contains primary amide group.

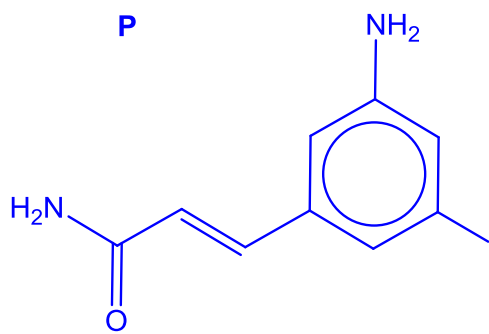
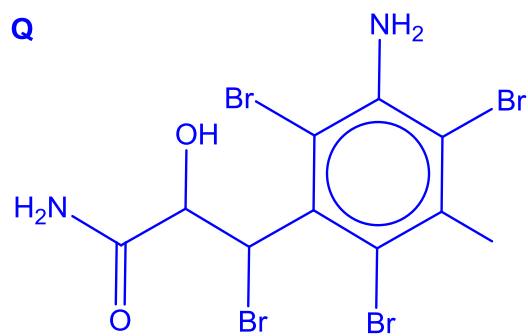
P undergoes both electrophilic substitution and electrophilic addition with aqueous bromine to form Q, $C_{10}H_{10}N_2O_2Br_4$ hence P contains both phenylamine and alkene group.

P undergoes oxidation with hot acidified $KMnO_4$ to form CO_2 gas hence P contains terminal alkene.



OR



OR**P****Q****R**